

Ratios And Rates Practice

mathNexa

Name:

Date:

Success Criteria:

- Write and simplify ratios from given quantities
- Calculate and compare unit rates in context
- Solve real-world multi-step rate problems

1. Look at the paint cans in the studio: there are 6 blue cans and 4 yellow cans. What is the ratio of blue cans to yellow cans in simplest form?



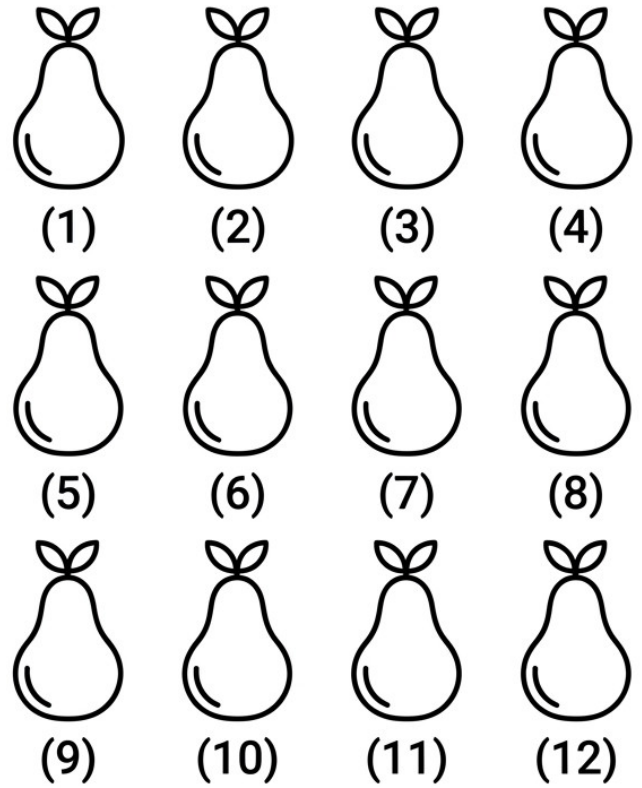
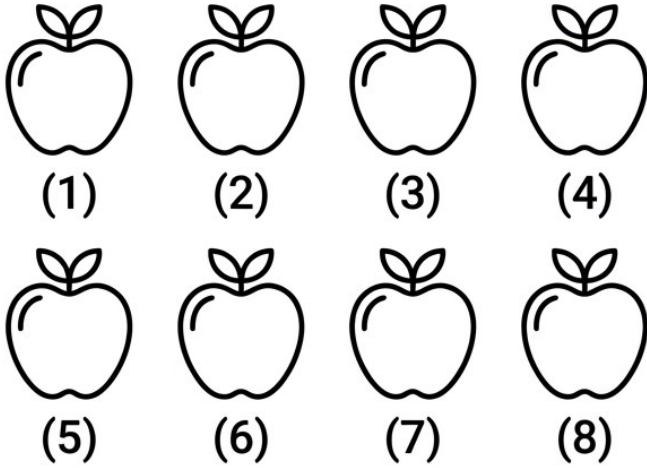
A 3 : 2

B 3 : 5

C 2 : 3

D 6 : 10

2. A fruit crate contains 8 red apples and 12 green pears. What is the ratio of red apples to the total number of fruits in simplest form?



Answer: _____

3. A lemonade recipe uses a ratio of 3 cups of lemon juice to 5 cups of water. Which ratio of lemon juice to water is equivalent?

A 6 : 8

B 12 : 16

C 5 : 3

D 9 : 15

4. A grocery store sells a pack of 6 identical bottles of orange juice for \$15.00 . What is the unit price per bottle in dollars?



Answer: _____

5. A cyclist travels 45 miles in 3 hours at a constant speed. What is the cyclist's unit rate of speed in miles per hour?

☐ A 15 mph ☐ B 12 mph ☐ C 45 mph ☐ D 18 mph

6. During exercise, Maya's heart beats 24 times in 15 seconds. At this rate, how many times will her heart beat in 60 seconds (1 minute)?

Answer: _____ beats

7. Brand A sells 18 ounces of granola for \$5.40 . Brand B sells 24 ounces of granola for \$6.72 . Which brand is the better buy per ounce, and what is its unit cost?

☐ A Brand B at \$0.28 per oz ☐ B Brand A at \$0.30 per oz
☐ C Brand B at \$0.32 per oz ☐ D Brand A at \$0.25 per oz

8. A car travels at a steady rate of 54 miles per hour on the highway. How many miles will the car travel in 2.5 hours?

Answer: _____ miles

9. A ribbon of length 48 cm is cut into two pieces in a ratio of 3 : 5 . What is the length of the longer piece of ribbon?

☐ A 40 cm

☐ B 30 cm

☐ C 18 cm

☐ D 24 cm

10. A printer prints 36 colour pages in 4 minutes. At this constant printing rate, how many minutes will it take to print a 135 -page report?

Answer: _____ minutes

Answers

1. A

4. 2.50

7. A

2. 2 : 5

5. A

8. 135

3. D

6. 96

9. B

10. 15